MODEL LAB · 2026-10-27

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)
served prior

11
priors compared

17,710
held-out plate appearances

2.92%
held-out HR rate

0.6014
best AUC · KNN comps

Held-out scores

Fitted on plate appearances before 2026-09-13, scored on the 17,710 that came after (actual HR rate 2.92%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5820	1.61×	0.02829	0.13108	0.00466
KNN comps Prior is the pooled rate of the nearest comparables.	0.6014	1.64×	0.02824	0.12993	0.00614
SVR Support-vector regression on the same feature matrix.	0.5940	1.51×	0.02829	0.13054	0.00689
Random forest Forest regression; a learned similarity metric.	0.5967	1.51×	0.02828	0.13022	0.00676
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6014	1.64×	0.02824	0.12993	0.00614
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6014	1.64×	0.02824	0.12993	0.00614
Linear regression Ordinary least squares on the same features.	0.5979	1.57×	0.02828	0.13031	0.00488
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5969	1.59×	0.02833	0.13060	0.00615
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5971	1.55×	0.02827	0.13021	0.00494
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5961	1.45×	0.02832	0.13060	0.00718

BLEND WEIGHTS (FOREST VARIANT)

KNN **1.00** SVR **0.00** RF **0.00**

SHRINKAGE K PER PRIOR

league **260** knn **260** svr **260** rf **260** core **260** ensemble **260** linear **260** logistic **260** nn **260** xgb **260**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (1,620 PA, actual HR rate 2.35%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.6456	1.58×	0.02291	0.11126	0.01790
KNN comps Prior is the pooled rate of the nearest comparables.	0.6062	1.58×	0.02288	0.10980	0.01007
SVR Support-vector regression on the same feature matrix.	0.5783	1.58×	0.02323	0.11613	0.02309
Random forest Forest regression; a learned similarity metric.	0.6000	1.32×	0.02320	0.11319	0.02260
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6062	1.58×	0.02288	0.10980	0.01007
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6062	1.58×	0.02288	0.10980	0.01007
Linear regression Ordinary least squares on the same features.	0.6112	1.32×	0.02322	0.11383	0.02025
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6015	1.32×	0.02354	0.11442	0.02384
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6056	1.58×	0.02300	0.11128	0.01557
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5934	1.05×	0.02361	0.11640	0.02762

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Jackson Chourio 17.1%, Junior Caminero 15.7%, William Contreras 15.4%, Jake Bauers 15.2%, Christian Yelich 13.2%, Jonathan Aranda 12.4%
Flat league	5/6	Jackson Chourio 17.4%, William Contreras 15.3%, Junior Caminero 15.0%, Jake Bauers 14.1%, Christian Yelich 13.3%, Yandy Díaz 12.5%
KNN comps	6/6	Jackson Chourio 17.7%, Junior Caminero 15.4%, William Contreras 15.4%, Jake Bauers 15.4%, Christian Yelich 13.5%, Jonathan Aranda 12.5%
SVR	5/6	Jackson Chourio 18.1%, Junior Caminero 16.8%, William Contreras 15.4%, Jake Bauers 15.1%, Christian Yelich 12.9%, Yandy Díaz 12.7%
Random forest	5/6	Jackson Chourio 18.0%, Junior Caminero 16.1%, Jake Bauers 15.4%, William Contreras 15.3%, Christian Yelich 13.2%, Yandy Díaz 12.5%
Core (KNN+SVR)	6/6	Jackson Chourio 17.7%, Junior Caminero 15.4%, William Contreras 15.4%, Jake Bauers 15.4%, Christian Yelich 13.5%, Jonathan Aranda 12.5%
Forest blend	6/6	Jackson Chourio 17.7%, Junior Caminero 15.4%, William Contreras 15.4%, Jake Bauers 15.4%, Christian Yelich 13.5%, Jonathan Aranda 12.5%
Linear regression	6/6	Jackson Chourio 17.9%, Junior Caminero 15.7%, William Contreras 15.4%, Jake Bauers 15.1%, Christian Yelich 13.5%, Jonathan Aranda 12.4%
Logistic regression	5/6	Jackson Chourio 18.0%, Junior Caminero 16.1%, William Contreras 15.4%, Jake Bauers 15.2%, Christian Yelich 12.9%, Yandy Díaz 12.3%
Neural net (torch)	5/6	Jackson Chourio 18.0%, Junior Caminero 16.0%, William Contreras 15.5%, Jake Bauers 15.3%, Christian Yelich 13.1%, Yandy Díaz 12.3%
XGBoost	5/6	Jackson Chourio 18.2%, Junior Caminero 16.3%, Jake Bauers 15.5%, William Contreras 15.3%, Christian Yelich 13.0%, Yandy Díaz 12.6%

Hitter by hitter

The top 40 bats by the served probability.

	HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
1	**Jackson Chourio** MIL Today	**17.1%**	17.4%	17.7%	18.1%	18.0%	17.7%	17.7%	17.9%	18.0%	
2	**Junior Caminero** TB Today	**15.7%**	15.0%	15.4%	16.8%	16.1%	15.4%	15.4%	15.7%	16.1%	
3	**William Contreras** MIL	**15.4%**	15.3%	15.4%	15.4%	15.3%	15.4%	15.4%	15.4%	15.4%	
4	**Jake Bauers** MIL	**15.2%**	14.1%	15.4%	15.1%	15.4%	15.4%	15.4%	15.1%	15.2%	
5	**Christian Yelich** MIL	**13.2%**	13.3%	13.5%	12.9%	13.2%	13.5%	13.5%	13.5%	12.9%	
6	**Jonathan Aranda** TB	**12.4%**	11.8%	12.5%	12.0%	12.1%	12.5%	12.5%	12.4%	12.2%	
7	**Yandy Díaz** TB	**12.2%**	12.5%	12.3%	12.7%	12.5%	12.3%	12.3%	12.4%	12.3%	
8	**Brice Turang** MIL	**11.9%**	12.0%	11.6%	11.8%	11.4%	11.6%	11.6%	11.1%	11.1%	
9	**Garrett Mitchell** MIL	**10.8%**	10.4%	10.7%	10.0%	10.4%	10.7%	10.7%	10.2%	10.2%	
10	**Ryan Vilade** TB	**10.7%**	10.0%	10.5%	10.2%	10.6%	10.5%	10.5%	10.2%	10.1%	
11	**Jonny DeLuca** TB	**8.3%**	8.9%	7.9%	7.8%	7.5%	7.9%	7.9%	7.9%	8.2%	
12	**Sal Frelick** MIL	**7.6%**	9.4%	7.8%	7.3%	7.0%	7.8%	7.8%	7.6%	7.7%	
13	**Liam Hicks** TB	**7.4%**	8.3%	7.3%	8.1%	7.9%	7.3%	7.3%	7.7%	7.6%	
14	**Cooper Pratt** MIL	**7.1%**	9.2%	7.0%	6.7%	6.5%	7.0%	7.0%	6.0%	6.4%	
15	**David Hamilton** MIL	**6.7%**	8.1%	6.5%	6.1%	5.9%	6.5%	6.5%	6.4%	6.4%	
16	**Richie Palacios** TB	**5.7%**	6.4%	5.7%	5.4%	5.0%	5.7%	5.7%	5.4%	5.4%	
17	**Taylor Walls** TB	**4.0%**	5.2%	3.8%	2.9%	2.9%	3.8%	3.8%	3.7%	3.8%	
18	**Chandler Simpson** TB	**2.1%**	4.8%	2.8%	1.3%	1.4%	2.8%	2.8%	1.2%	2.5%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

Slate for 2026-10-27 · model 2026.10.01-stack · built 2026-10-28 05:31:49 · live data 2026-10-28 05:31:49

Built for research and entertainment. Probabilities are estimates, not guarantees; never stake what you cannot afford to lose. Data: MLB Stats API play-by-play, lineups and live scores; ESPN injuries and bios.